PAVAN KUMAR GUMMULURU

AI/ML ENGINEER | DATA SCIENTIST | HEALTHCARE & GENERATIVE AI SPECIALIST

MASSACHUSETTS, USA | 518-400-2740 | PAVANKUMARGUMMULURU@GMAIL.COM |LINKEDIN

PROFESSIONAL SUMMARY

AI & ML engineer with 4 years of experience in designing, developing, and deploying data-driven solutions across healthcare, e-commerce, finance, and manufacturing domains. Skilled in machine learning, deep learning (CNN, RNN, Transformers), NLP (LLMs, BERT, GPT), and computer vision (yolo, OpenCV) for solving business-critical problems. Proficient in MLOps with MLflow, TensorFlow Serving, Torch Serve, Docker, Kubernetes, and CI/CD pipelines, ensuring scalable and production-ready deployments on AWS SageMaker, GCP Vertex AI, and Azure ML. Experienced in building recommendation systems, fraud detection models, medical imaging pipelines, and clinical NLP solutions, driving measurable improvements in accuracy, efficiency, and compliance. Adept at data engineering workflows (Spark, Airflow, SQL), generative AI (LangChain, Rag Systems, Vector Databases), and effective stakeholder communication, bridging the gap between business needs and AI-driven innovation.

CORE SKILLS

Machine learning: regression, classification, clustering, ensemble methods (XGBoost, random forest), model evaluation

Deep learning: CNN, RNN/LSTM, transformers

Natural language processing (NLP): LLMS (GPT, Bert), Hugging Face, Prompt Engineering, Rag Systems

Computer vision: object detection (yolo), image segmentation, OpenCV

MLOps deployment: MLflow, TensorFlow Serving, torch serve, Docker, Kubernetes, CI/CD

Generative AI: fine-tuning LLMs, LangChain, vector databases (Faiss, Pinecone)

Data engineering: Apache Spark, Airflow, SQL

Programming languages: Python (pandas, PyTorch, TensorFlow, scikit-learn), SQL, C++

Cloud platforms: AWS, Azure

Tools & version control: git/github, Jenkins, Jira

Business & communication: stakeholder communication, problem framing, technical documentation

PROFESSIONAL EXPERIENCE

CARDINAL HEALTH – MASSACHUSETTS, USA

AI ENGINEER| *DEC 2024 – PRESENT*

- Designed and deployed predictive models to forecast patient readmissions and medication adherence using XGBoost, random forest, and LSTM networks, improving prediction accuracy by 18%.
- Developed computer vision pipelines with yolo and OpenCV for medical inventory tracking and anomaly detection in radiology scans, reducing manual review time by 25%.
- Implemented NLP-driven clinical note analysis leveraging BERT/GPT models and RAG systems to extract key medical entities and automate documentation, saving 12+ hours/week for physicians.
- Deployed scalable AI models with MLflow, Docker, and Kubernetes integrated into AWS SageMaker for real-time healthcare workflows, ensuring compliance with HIPAA standards.
- Collaborated with cross-functional teams to frame business problems, translate them into ML-driven healthcare solutions, and communicate insights effectively to clinical stakeholders.
- Built end-to-end CI/CD pipelines for ML models, integrating TensorFlow Serving and Torch Serve, ensuring continuous delivery of updated models into production.

ORION TECHNOLAB – INDIA

ML ENGINEER| *JAN 2020 – JULY 2023*

- Developed recommendation systems for e-commerce clients using collaborative filtering, matrix factorization, and deep learning, boosting click-through rates by 22%.
- Implemented fraud detection models for financial clients using ensemble methods (XGBoost, LightGBM), reducing false positives by 15%.
- Built customer sentiment analysis pipelines with Hugging Face transformers to analyze social media and support tickets, enabling proactive client engagement.
- Created image classification and defect detection models with CNNs and OpenCV for manufacturing clients, reducing quality control errors by 30%.
- Automated ETL workflows with Apache Spark and Airflow, improving data pipeline efficiency and reducing processing time by 40%.
- Deployed ML solutions using GCP Vertex AI and Azure ML, ensuring scalability across client infrastructures.
- Documented technical solutions, mentored junior engineers, and collaborated with business teams to align ML solutions with client KPIs.

ACADEMIC PROJECTS

CONTEXT-AWARE IMAGE CAPTIONING USING LLM + VLM

UNIVERSITY OF MASSACHUSETTS DARTMOUTH

- Developed a hybrid LLM-VLM framework using pseudo-labeling and prompt optimization to improve image caption accuracy on the Whoops dataset (500 samples).
- Integrated Bert embeddings and iterative feedback for semantic refinement.
- Evaluated captions using BLEU scores and ensured reproducibility in Jupyter Lab with Python-based pipelines.

CAR PRICE PREDICTION

UNIVERSITY OF MASSACHUSETTS DARTMOUTH

- Built a regression model in scikit-learn to predict car prices using technical specs and body features.
- Performed EDA to uncover pricing drivers and trends.
- Designed Tableau dashboards to visualize pricing insights and recommendations.

EDUCATION

MASTER'S IN DATA SCIENCE

UNIVERSITY OF MASSACHUSETTS DARTMOUTH | *MAY 2025*

CERTIFICATIONS

- **TABLEAU FOR DATA SCIENCE** – COURSERA
- **ADVANCED SQL FOR DATA ANALYTICS** – UDEMY
- **POWER BI DASHBOARDING** – LINKEDIN LEARNING